Assignment 2

Date
Page

	0	1	2	3	4
0	*	#	*	#	*
1	#	*	#	*	#
2	*	#	*	#	*
3	#	*	#	*	#
4	*	#	*	#	*

1)

0+0%=25
0+2
0+4

```
int n=5
for(int i=0; i<n; i++){
    for(int j=0; j<n; j++){
        if(i+j%2==0){
            sop("* ");
        }
        else{
            sop("# ");
        }
    }
}
```

int i=0 0<5 ①
└ j=0 0<5 ①
└ i+j%2
0%2==0
└ *

j=1 1<5
└ 0+1%2==0
└ #

j=2 2<5 ③

0+2 %2==0

*

j=3 3<5

0+3

3%2!=0

#

j=4 4<5

0+4

4%2==0

*

j=5 5<5 ✗

*

i=1 1<5 ②

j=0 0<5

0+1

1%2!=0

#

j=1 1<5

1+1

2%2==0

*

j=2 2<5

└ 1+2

3%2 != 0

└ #

j=3 3<5

└ 1+3

4%2 == 0

└ *

j=4 4<5

└ 1+4

5%2 != 0

└ #

j=5 5<5 └ X

i=2 2<5 ①

└ j=0 0<5

└ 2+0

2%2 == 0

└ *

j=1 1<5

└ 2+1

3%2 != 0

└ #

j = 2 2 < 5

2 + 2

4 % 2 = = 0

*

j = 3 3 < 5

3 + 2

5 % 2 ! = 0

#

j = 4 4 < 5

4 + 2

6 % 2 = = 0

*

j = 5 5 < 5 ✗

2)

	0	1	2	3	4
0	*	*	*	*	*
1	*				*
2	*				*
3	*				*
4	*	*	*	*	*

```
int n=5;
for(int i=0 ; i<n; i++) {
        for(int j=0;j<n; j++ ){
                if(i==0 || i==n-1 || j==0 || j==n-1)
                        S.OP("*");
                }
                else {
                        S.O.P(" ");
                }
        }
    SOP();
}.
```

int i=0 0 < 5 i++ $\boxed{1}$

└ j=0 j < 5 ①

└ i==0

└ *

j=1 └ 1<5

└ i==0

└ *

j = 2 2 < 5
└ i == 0
└ *

j = 3 3 < 5
└ i == 0
└ *

j = 4 4 < 5
└ i == 0
└ *

j = 5 5 < 5 └ ×

j = 2 2 < 5 (3)
└ j = 0 0 < 5
└ j == 0
└ *

j = 1 1 < 5
└ i != 0
i != 4
j != 0
j != 4
└ " "

j = 2 2 < 5
└ i != 0
i != 4
j != 0
j != 4
└ " "

j=3 3<5

" "

j=4 4<5

j==4

*

j=5 5<5 ×

i=3 3<5 ④

j=0 0<5

i!=0

j==0

*

j=1 1<5

i!=0

i!=4

j!=0

j!=4

" "

j=2 2<5

i!=0

i!=4

j!=0

j!=4

" "

j=3 3<5

" "

j = 4 4 < 5

j == 4

*

j = 5 5 < 5 ×

3)

	0	1	2	3	4
4	*	*	*	*	*
3	*	*	*	*	
2	*	*	*		
1	*	*			
0	*				

4<0

```
int n=5
for (int i=n-1; j>=0; j--) {
        for(int j=0; j<=i; j++) {
                s.op(" * ");
        }
        sop();
}
}
```

int i = 4 4 >= 0 3

└ j = 0 j <= i j++

0 0 <= 4

└ *

1 1 <= 4

└ *

2 2 <= 4

└ *

3 3 <= 4

└ *

4 4 <= 4

└ *

5 5 <= 4 ✗

i = 3 3 >= 0 2

└ j = 0 0 <= 3

└ *

j = 1 1 <= 3

└ *

j=2 2<=3
*

j=3 3<=3
*

i=2 2>=0 ①
j=0 0<=2
*

j=1 1<=2
*

j=2 2<=2
*

j=3 3<=2
×

i=1 1>=0
j=0 0<=1
*

j=1 1<=1
*

j=2 2<=1
×

4)

	0	1	2	3	4
0	*				
1	*	*			
2	*	*	*		
3	*	*	*	*	
4	*	*	*	*	*

```
int n=5
for (int i=0; i<n; i++) {
        for (int j=0 ; j<=i ; j++) {
                sop("*");
        }
        sop().
    }
}
```

int i=0 0<5 ①

└ j=0 j<=i j++

0<=0

└ *

j=1 1<=0

└ ×

i=1 1<5

j = 0 0 <= 1
└ *

j = 1 1 <= 1
└ *

j = 2 2 <= 1
└ X

i = 2 2 < 5 ③
└ j = 0 0 <= 2
└ *

j = 1 1 <= 2
└ *

j = 2 2 <= 2
└ *

j = 3 3 <= 2
└ X

i = 3 3 < 5
└ j = 0 0 <= 3
└ *

j = 1 1 <= 3
└ *

j=2 2<=3
└ *

j=3 3<=3
└ *

j=4 4<=3
└ X

i=4 4<5
└ j=0 0<=4
└ *

j=1 1<=4
└ *

j=2 2<=4
└ *

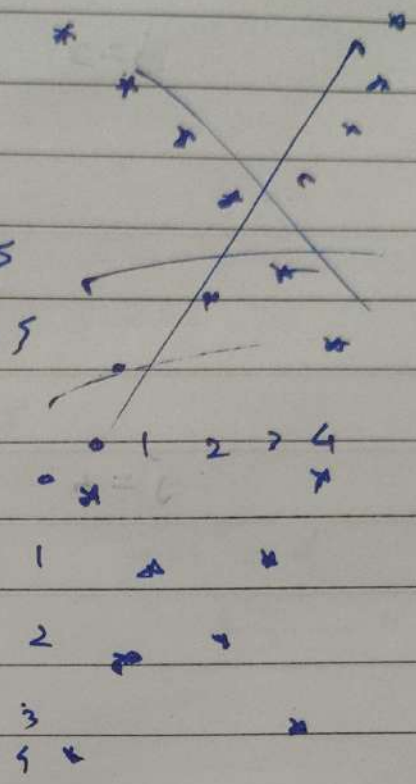
j=3 3<=4
└ *

j=4 4<=4
└ *

j=5 5<=4
└ X

5) int n=5;

```
for(int i=0; i<n; i++){
    for(int j=0; j<n; j++){
        if(i==j||(i+j==n-1)){
            sop("*")
        }
    else {
        sop(" ").
```



int i=0 0<5 [1]

└ j=0 0<5

└ i==j

└ *

j=1 1<5

└ i!=j

0+1 != 4

└ " "

j=2 2<5

└ i!=j

0+2 != 4

└ " "

j=3 3<5

j!=j

i+j

0+3!=4

" "

j=4 4<5

i!=j

0+4==4

*

j=5 5<5 ×

i=1 1<5

j=0 0<5 ②

j!=i

0+1!=4

" "

j=1 1<5

i==j

*

j=2 2<5

i!=j

1+2!=4

" "

j=3 3<5

i!=j

1+3==4

*

j=4 4<5

i!=j

4+1!=4

" "

j=5 5<5 ×

j=2 2<5

j=0 0<5

i!=j

0+2!=4

" "

j=1 1<5

i!=j

1+2!=4

" "

j=2 2<5

j==j

*

j=3 3<5

i!=j

3+2 != 4

" "

j=4 4<5

i!=j

4+3 != 4

" "

j=5 5<5 ✗

i=3 3<5 ④

j=0 0<5

i!=j

0+3 != 4

" "

j=1 1<5

i!=j

1+3 == 4

✱

j=2 2<5

i!=j

2+3 != 4

" "

j = 3 3 < 5

└ i == j

└ *

j = 4 4 < 5

└ i != j

3+4 != 4

└ " "

j = 5 5 < 5

└ X